

The Human Cost of Respiratory Care

Professional Quality of Life and Burnout Across Multidisciplinary Cystic Fibrosis Teams

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BACKGROUND: Burnout is characterized by emotional exhaustion, reduced personal accomplishment, and increased depersonalization. Over one-half of physicians, nurses, and allied health professionals report substantial burnout, with consequences including decreased empathy and quality of care, increased medical errors, and increased health care costs. Team-based care, common in chronic respiratory conditions such as cystic fibrosis (CF), may be protective.

RESEARCH QUESTION: What is the risk of burnout among multidisciplinary CF care teams and does it differ by profession?

STUDY DESIGN AND METHODS: US CF health care providers received a 64-item survey including 2 standardized measures (Professional Quality of Life Scale and Maslach Burnout Inventory). Respondents (N = 569) included physicians (24.9%), advanced practice providers (APPs)/nurses (21.9%), allied health (29.4%) and mental health providers (23.8%).

RESULTS: Providers reported moderate compassion satisfaction and low personal accomplishment, with low-moderate burnout and moderate emotional exhaustion and depersonalization. Significant differences were found by profession ($F_{3,585} = 3.77-5.07$; $P < .05-.01$): in comparison with mental health providers, physicians, APPs/nurses, and allied health providers reported lower personal accomplishment and higher burnout; physicians and allied health providers reported higher depersonalization; and physicians reported higher emotional exhaustion. Overall, women reported lower personal accomplishment than men. Hispanic providers reported higher burnout and emotional exhaustion and lower personal accomplishment. Providers working in CF ≥ 15 years reported less burnout than those working 10 to 14 years. Top contributors to burnout were the following: administrative tasks (80.0%), excessive documentation (79.4%), insufficient staffing (74.6%), insufficient time with patients (73.4%), and excessive responsibility (72.7%). One-half of mental health providers had sought professional help for burnout vs only one-third of physicians; significantly more female physicians sought help (40%) than male physicians (19%).

INTERPRETATION: Our results show that despite well-established, team-based models of pulmonary care, burnout remained substantial, with physicians at highest risk, followed by APPs/nurses and allied health providers. In addition to promoting individual well-being, systems-level initiatives are critical to support multidisciplinary teams caring for people with chronic pulmonary disease.

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KEY WORDS: burnout; cystic fibrosis teams; depersonalization; health care providers; professional quality of life; team-based care

Take-Home Points

Research Question: What is the risk of burnout among multidisciplinary cystic fibrosis care teams and does it differ by profession?

Results: Overall, cystic fibrosis health care providers reported moderate burnout, emotional exhaustion, and depersonalization with low personal accomplishment. In comparison with mental health providers, physicians, nurses, and allied health providers reported more burnout and less personal accomplishment.

Interpretation: Despite well-established, team-based models of pulmonary care, burnout remained substantial, with physicians at highest risk, followed by advanced practice providers/nurses and allied health providers.

Burnout is a work-related syndrome characterized by emotional exhaustion, reduced personal accomplishment and fulfillment, and increased depersonalization.¹ Over one-half of physicians, nurses, and allied health professionals (ie, dietitians, respiratory therapists, physical therapists, pharmacists, medical assistants, pharmacy technicians) report substantial burnout.^{2,3} Pulmonologists consistently fall in the top tier of burnout risk,⁴ ranked fourth among medical specialties.^{5,6} Burnout is associated with decreased empathy and quality of care, higher rates of medical errors, and costs to the health care system,⁷ including > \$5 billion yearly due to physician turnover.⁸⁻¹¹ The COVID-19 pandemic led to increased provider stress and burnout in the context of increased workload with perceived lack of control.¹²⁻¹⁴ Rates of burnout were particularly high for pulmonologists who worked in intensive care settings, peaking in 2022 at 68.4%.⁴

Individual risk factors for burnout include gender, minoritized status, and length of professional

practice.^{2,15,16} Female physicians typically report higher levels of burnout than their male counterparts because they face a higher workload, spend more time with patients and documentation, and report less control over their schedules.¹⁷⁻¹⁹ At a systems level, women have lower rates of career advancement, including leadership roles, and disproportionate responsibilities (eg, family roles) outside of work.¹⁷⁻¹⁹ Differences by minority status are less clear. A systematic review showed higher rates of burnout in underrepresented minoritized populations, whereas others showed no differences; however, few studies have evaluated levels of racism or implicit bias.²⁰ In a longitudinal cohort study with high response rates, physicians with fewer years of experience were at risk for higher burnout.¹⁵

One critical factor that may mitigate burnout is the practice of team-based care,²¹ common in treating chronic pulmonary conditions such as cystic fibrosis (CF), asthma, and COPD.²²⁻²⁴ Team-based care reduces burnout by decreasing workload, improving workflow, fostering social support, and facilitating collaborative care.^{24,25} Team-based care also leads to better health outcomes for patients including better adherence, reduced hospitalizations and emergency visits, and better quality of life (QOL), all of which may indirectly reduce provider burnout.²¹ In a study of > 1,200 physicians, ratings of teamwork climate and safety were associated with lower scores on the Maslach Burnout Inventory (MBI).^{21,26} For each 1-point increase in teamwork, the odds of high emotional exhaustion were reduced 16% and the odds of high depersonalization were reduced by 26%.²¹

The objective of this study was to evaluate professional QOL and burnout using validated measures in multidisciplinary CF care teams, which have a long-standing tradition of team-based care for pediatric and adult patients.²³ To our knowledge, this is the first national study of professional QOL and burnout in

ABBREVIATIONS: APP = advanced practice provider; CF = cystic fibrosis; MBI = Maslach Burnout Inventory; ProQOL = Professional Quality of Life Scale; QOL = quality of life

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providers working on pulmonary multidisciplinary teams. We expected CF health care providers to report high levels of professional QOL on measures of compassion satisfaction and personal accomplishment, and moderate levels of burnout and depersonalization. We anticipated that providers in higher status/higher salaried positions with greater autonomy and control (eg, physicians) would report higher satisfaction and personal accomplishment and lower levels of burnout than other members of the clinical care team.

Additionally, we evaluated potential relationships between perceptions of professional QOL and burnout,

expecting that those endorsing high professional QOL would report lower levels of burnout. We examined potential predictors of burnout, hypothesizing that women would report lower personal accomplishment and higher burnout than men, minoritized providers would report higher burnout, and those with more years working in CF would report more positive professional QOL and less burnout. We expected professional burden (eg, documentation, staffing shortages) to be perceived as contributors to burnout, and asked respondents to rank order coping strategies they used to mitigate burnout.

Study Design and Methods

Design

This was a cross-sectional health care provider survey, including standardized measures of professional QOL and burnout. Respondents were assured that their responses would be anonymous and confidential.

Survey and Measures

The survey (see e-Appendix 1) included the following 4 sections: (1) demographic and professional information (eg, age, race/ethnicity, profession, years on a CF team), (2) Professional Quality of Life Scale (ProQOL),²⁷ (3) MBI,^{1,26} and (4) rank ordering contributors to and coping strategies for burnout.

The CF Foundation Mental Health Advisory Committee—including 7 CF providers (eg, psychologists, psychiatrist, pulmonologist, social worker), 1 community advisor, and 1 CFF staff member—selected measures and developed the survey through several iterations, adhering whenever possible to the Dillman Tailored Design Method and STROBE checklist.²⁸⁻³⁰ Survey questions were customized to CF providers and piloted in a small study.³¹

The ProQOL version 5³² evaluates positive and negative aspects of professionals' QOL: 10-item compassion satisfaction and 20-item compassion fatigue, which includes 2 scales: 10-item burnout and 10-item secondary traumatic stress (30 questions in total). For this study, we analyzed the 2 scales that make-up compassion fatigue, burnout, and secondary traumatic stress, separately. Each item is rated for frequency over the past 30 days on a 5-point scale, taking 8 minutes to complete. All ProQOL scores are divided into low (≤ 22), moderate (23-41), and high (≥ 42). Each scale is independent with strong reliability and validity.³² In the

current study, ProQOL subscales demonstrated adequate reliability (compassion satisfaction: $\alpha = 0.91$; burnout: $\alpha = 0.83$; secondary traumatic stress: $\alpha = 0.82$).

The MBI^{1,26} is the most widely used and is the gold standard for assessing burnout in health care providers. The MBI has 3 scales: 9-item emotional exhaustion, 8-item personal accomplishment, and 5-item depersonalization. Items are rated on a 6-point scale from never (0) to every day (6), taking 10 minutes to complete. The MBI emotional exhaustion scores are divided into low (≤ 17), moderate (18-29), and high (≥ 30). The MBI personal accomplishment scores are divided into low (≤ 33), moderate (34-39), and high (≥ 40). The MBI depersonalization scores are divided into low (≤ 5), moderate (6-11), and high (≥ 12). It has excellent internal consistency and test-retest reliability over 1 month and 1 year and good convergent and discriminant validity.^{26,33} In this study, MBI subscales demonstrated strong reliability (MBI emotional exhaustion: $\alpha = 0.93$; MBI personal accomplishment: $\alpha = 0.83$; MBI depersonalization: $\alpha = 0.80$).

Survey Dissemination

The CF Foundation distributed a REDCap link to members of multidisciplinary CF teams in the US via email and listservs, with 2 email reminders; the survey was open from May 30, 2024, to August 4, 2024. The introduction described the prevalence of health care provider burnout and its consequences, with a consent statement and assurance that responses were deidentified and confidential. All respondents received links to a CF Foundation Mental Health Advisory Committee tips sheet on burnout and self-care tools (proqol.org/self-care-tools). Incentives were not provided. Institutional Review Boards at Mass General Brigham

(No. ID1792) and Memorial Healthcare System determined that the project did not meet criteria for human patients research.

Analyses

Descriptive statistics (eg, means, SDs, quintiles, frequencies, percentages) were calculated. After analyzing missing data, we used a complete case analysis for participants who completed the primary outcome (ProQOL): *t* tests for independent means, analyses of variance with Bonferroni post hoc tests, and Spearman correlations analyzed relationships between provider characteristics and continuous study outcomes (ProQOL, MBI emotional exhaustion, and MBI personal accomplishment). ORs and 95% CIs evaluated

Results

Study Flow

The email was opened by 1,047 recipients with 687 unique participants opening the survey link, generating a REDCap study ID, and 645 who completed at least 1 item. Of these, 612 served patients in a clinical role; 33 endorsed nonclinical roles (eg, research coordinator) and were removed from analyses. Of the 612 participants with clinical roles, 569 (93%) completed the ProQOL, which served as the analytical sample.

Provider Professional Roles and Demographic Characteristics

Respondents were physicians (24.9%), advanced practice providers (APPs)/nurses (21.9%), allied health providers (eg, respiratory therapists, dietitians) (29.4%), and mental health providers (ie, social workers, psychologists, psychiatrists) (23.8%) (Table 1). One-quarter were in their 30s, 40s, and 50s, respectively. Most identified as White (83.5%), non-Hispanic (91.3%), and female (81.9%) participants. Among physicians, however, 52% were female. Almost one-half of the respondents worked with pediatric patients (48.3%) in large programs (42.8%). One-third worked in CF $\geq$ 15 years.

Overall ProQOL and MBI Scores

On average, on the ProQOL, participants scored in the moderate range on the compassion satisfaction scale (mean $\pm$ SD, 39.8 ± 6.1), in the low to moderate range for burnout (mean $\pm$ SD, 22.4 ± 5.7), and in the low range for secondary traumatic stress (mean $\pm$ SD, 20.4 ± 5.4) (Table 2).

associations between provider characteristics and high/moderate MBI depersonalization scores. General linear models (ProQOL, MBI emotional exhaustion, and MBI personal accomplishment) and logistic regression (MBI depersonalization) evaluated relationships between professional role and outcome variable significant in bivariate analyses, adjusting for significant covariates. For the 5 models with linear outcomes, we used GLMs with a Gaussian distribution and an identity link. For the one model with a dichotomous outcome (MBI depersonalization), we used GLMs with a binomial distribution and a logit link. SPSS version 27 was used for the analyses. All inferential statistics were 2-sided with $P = .05$.

On the MBI, burnout is typically characterized by high emotional exhaustion, moderate depersonalization, and low personal accomplishment. Our analysis found that, on average, participants scored in the moderate range on MBI emotional exhaustion (mean $\pm$ SD, 21.0 ± 12.5) and MBI depersonalization (mean $\pm$ SD, 6.2 ± 5.6), and low on MBI personal accomplishment (mean $\pm$ SD, 31.8 ± 8.4) (Table 2). Due to the nonnormal distribution of depersonalization, this variable was dichotomized into low vs moderate/high levels for the remaining analyses.

Bivariate Associations Between Professional Roles and ProQOL and MBI Scores

Significant differences were found across most scales by profession. Bivariate relationships between professional role and ProQOL and MBI are reported in Figure 1 and Table 2. Personal accomplishment differed across professional roles ($F_{3,585} = 5.32$; $P = .001$). Compared with mental health providers, physicians (-2.7 ; 95% CI, -5.4 to -0.1), APPs/nurses (mean difference, -3.5 ; 95% CI, -6.3 to -0.7), and allied health providers (-3.5 ; 95% CI, -6.1 to -0.9) reported significantly lower personal accomplishment. No differences in compassion satisfaction were found among professional roles.

ProQOL burnout also differed by professional roles ($F_{3,585} = 5.07$; $P = .002$). Compared with mental health providers, physicians (mean difference, 2.8 ; 95% CI, 0.1 - 5.5), APPs/nurses (mean difference, 3.5 ; 95% CI, 0.7 - 6.3), and allied health providers (mean difference, 3.5 ; 95% CI, 0.9 - 6.1) all reported significantly higher burnout (Fig 1, Table 2).

TABLE 1] Provider Characteristics and Professional Roles (N = 569)

Characteristic	No. (%)
Demographic	
Age, y (n = 562)	
20-30	31 (5.5)
31-40	155 (27.6)
41-50	147 (26.2)
51-60	144 (25.6)
≥ 61	85 (15.1)
Race (nonexclusive)	
White	475 (83.5)
Asian	43 (7.6)
Black/African American	16 (2.8)
American Indian	3 (0.5)
Native Hawaiian	1 (0.2)
Another race	12 (2.1)
Multiracial	11 (1.9)
Prefer not to say	10 (1.8)
Ethnicity (n = 561)	
Hispanic/Latino	49 (8.7)
Non-Hispanic/Latino	512 (91.3)
Sex (n = 559)	
Male	98 (17.5)
Female	458 (81.9)
Other	3 (0.5)
Gender (n = 560)	
Man	96 (16.9)
Woman	458 (80.5)
Gender nonbinary or genderqueer	6 (1.1)
Professional	
Population (n = 563)	
Pediatric	272 (48.3)
Adult	168 (29.8)
Both pediatric and adult	123 (21.8)
Program size (n = 568)	
Small (> 100)	112 (19.7)
Medium (100-200)	213 (37.5)
Large (> 200)	243 (42.8)
Length of time working with CF, y (n = 567)	
< 1	27 (4.8)
1-4	116 (20.5)
5-9	128 (22.6)
10-14	112 (19.8)
≥ 15	184 (32.5)

(Continued)

TABLE 1] (Continued)

Characteristic	No. (%)
Professional role (n = 562)^a	
Physician, nonpsychiatrist	140 (24.9)
Advanced practice provider/nursing	123 (21.9)
physician assistant	8 (6.5)
Advanced nurse practitioner	11 (8.9)
Nurse practitioner	24 (19.5)
Nurse, other	80 (65.0)
Allied health providers	165 (29.4)
Dietician	60 (36.4)
Respiratory therapist	53 (32.2)
Physical therapist	12 (7.3)
Pharmacist	31 (18.8)
Medical assistant or pharmacy technician	9 (5.4)
Mental health providers	134 (23.8)
Psychiatrist	2 (1.5)
Psychologist	23 (17.2)
Social worker	104 (77.6)
Mental health counselor	3 (2.2)
Child life specialist	2 (1.5)

CF = cystic fibrosis.

^aNo. (%) for subcategories under each professional role indicate the % of the professional role category that endorsed each subcategory.

MBI emotional exhaustion also differed by profession ($F_{3,585} = 3.77$; $P = .011$). Physicians reported significantly higher emotional exhaustion than mental health providers (mean difference, 4.9; 95% CI, 0.9-9.0). Odds of reporting moderate/high scores on the MBI depersonalization also differed by professional roles ($\chi^2_3 = 10.8$; $P = .013$). Compared with mental health providers, odds of reporting moderate/high depersonalization were higher for physicians (OR, 2.2; 95% CI, 1.4-3.7) and allied health providers (OR, 1.9; 95% CI, 1.1-3.0) (Fig 1, Table 2).

Comparisons of Mental Health Providers: Subgroup sensitivity analyses revealed significant differences between psychiatry/psychology (n = 25) and social work (n = 104) on ProQOL burnout ($t_{127} = -2.67$; $P = .010$), ProQOL secondary traumatic stress ($t_{127} = -3.34$; $P = .002$), and MBI personal accomplishment ($t_{123} = 4.17$; $P < .001$). Psychiatrists/psychologists reported significantly lower burnout (mean $\pm$ SD, 19.1 ± 3.7) than social workers (mean $\pm$ SD, 21.5 ± 5.2 ; mean difference, -2.4 ; 95% CI, -4.2 to -0.6) and significantly lower secondary traumatic stress (mean $\pm$ SD, 17.4 ± 4.0)

TABLE 2] Average ProQOL and MBI Scores Across Providers and Bivariate Associations Between Professional Role and ProQOL and MBI

	ProQOL			MBI		
	CS	BO	STS	EE	PA	DP
Overall score	(N = 569) 39.8 [6.1]	(N = 569) 22.4 [5.7]	(N = 569) 20.4 [5.4]	(n = 539) 21.0 [12.5]	(n = 538) 31.8 [8.4]	(n = 537) 6.2 [5.6]
Professional Role	(n = 592)	(n = 592)	(n = 592)	(n = 559)	(n = 557)	DP Moderate/High (n = 558) No./Total No. (%) ^a
Physician, nonpsychiatrist (n = 140)	39.1 [5.9]	23.6 [5.9]	20.8 [5.0]	22.9 [13.1]	31.6 [8.4]	68/134 (50.7)
Advanced practice provider and nursing (n = 123)	40.1 [6.6]	22.9 [5.9]	20.9 [5.3]	21.7 [12.6]	30.8 [8.3]	47/113 (41.6)
Allied health providers (n = 165)	39.8 [5.6]	22.2 [5.6]	20.1 [5.7]	21.4 [13.3]	30.8 [8.2]	71/154 (46.1)
Mental health providers (n = 134)	40.6 [6.1]	21.1 [5.0] ^b	19.9 [5.4]	18.0 [10.5] ^b	34.2 [8.2] ^b	41/130 (31.5) ^b

Values are presented as mean [SD] or as otherwise indicated. Boldface values are significantly different from mental health providers (reference). Post hoc tests were corrected using the Bonferroni correction. Scoring criteria included the following: ProQOL (CS, BO, and STS) (low: ≤ 22 ; moderate: 23-41; high: ≥ 42); MBI EE (low: ≤ 17 ; moderate: 18-29; high: ≥ 30); MBI PA (low: ≤ 33 ; moderate: 34-39; high: ≥ 40); and MBI DP (low: ≤ 5 ; moderate: 6-11; high: ≥ 12). BO = burnout; CS = compassion satisfaction; DP = depersonalization; EE = emotional exhaustion; PA = personal accomplishment; ProQOL = Professional Quality of Life Scale; STS = secondary traumatic stress.

^aDP is summarized as the moderate and high categories.
^bReference.

than social workers (mean $\pm$ SD, 20.6 $\pm$ 5.4; mean difference, -3.2; 95% CI, -5.2 to -1.3). Psychiatrists/psychologists reported significantly higher MBI personal accomplishment (mean $\pm$ SD, 38.6 $\pm$ 4.6) than social workers (mean $\pm$ SD, 33.2 $\pm$ 8.7; mean difference, 5.3; 95% CI, 2.8-7.9). No other significant differences on ProQOL and MBI were found.

Bivariate Associations Between Provider Characteristics and Outcomes

Bivariate associations between provider characteristics and outcomes are found in e-Table 1. Higher age (assessed by decade) was associated with higher ProQOL compassion satisfaction ($p = 0.14$; $P < .001$), lower ProQOL burnout ($p = -0.14$; $P = .001$), lower MBI emotional exhaustion ($p = -0.15$; $P < .001$), higher MBI personal accomplishment ($p = 0.09$; $P = .032$), and lower MBI depersonalization ($p = -0.15$; $P < .001$). As hypothesized, gender was associated with MBI personal accomplishment ($t_{522} = 2.40$; $P = .018$), with men reporting higher personal achievement than women (mean difference, 2.3; 95% CI, 0.4-4.2). Ethnicity was associated with ProQOL burnout ($t_{559} = 2.40$; $P = .017$), ProQOL secondary traumatic stress ($t_{559} = 3.4$; $P < .001$), MBI emotional exhaustion ($t_{529} = 2.3$; $P = .022$), and MBI personal accomplishment ($t_{527} = -2.00$; $P = .049$). In all cases, participants who were Hispanic/Latino reported worse outcomes than those who were non-Hispanic/Latino.

Length of Time Working With CF teams: As expected, providers who had worked ≥ 15 years on CF teams reported better outcomes on ProQOL compassion satisfaction ($F_{2,564} = 8.10$; $P < .001$), ProQOL burnout ($F_{2,564} = 7.95$; $P < .001$), MBI emotional exhaustion ($F_{2,534} = 6.31$; $P = .002$), and MBI personal accomplishment ($F_{2,532} = 3.35$; $P = .036$) (e-Table 1, Fig 2). Participants working in CF for < 10 or ≥ 15 years reported higher compassion satisfaction than those working in CF 10 to 14 years (mean difference, 1.9; 95% CI, 0.3-3.5; mean difference, 2.9; 95% CI, 1.2-4.6); they also reported lower burnout on the ProQOL (mean difference, -2.2; 95% CI, -3.7 to -0.7; mean difference, -2.5; 95% CI, -4.1 to -0.9) and lower emotional exhaustion on the MBI (mean difference, -5.0; 95% CI, -8.4 to -1.6; mean difference, -4.3; 95% CI, -8.0 to -0.6). Participants working with CF teams for ≥ 15 years reported higher MBI personal accomplishment than those working with CF teams for 10 to 14 years (mean difference, 2.6; 95% CI, 0.1-5.2).

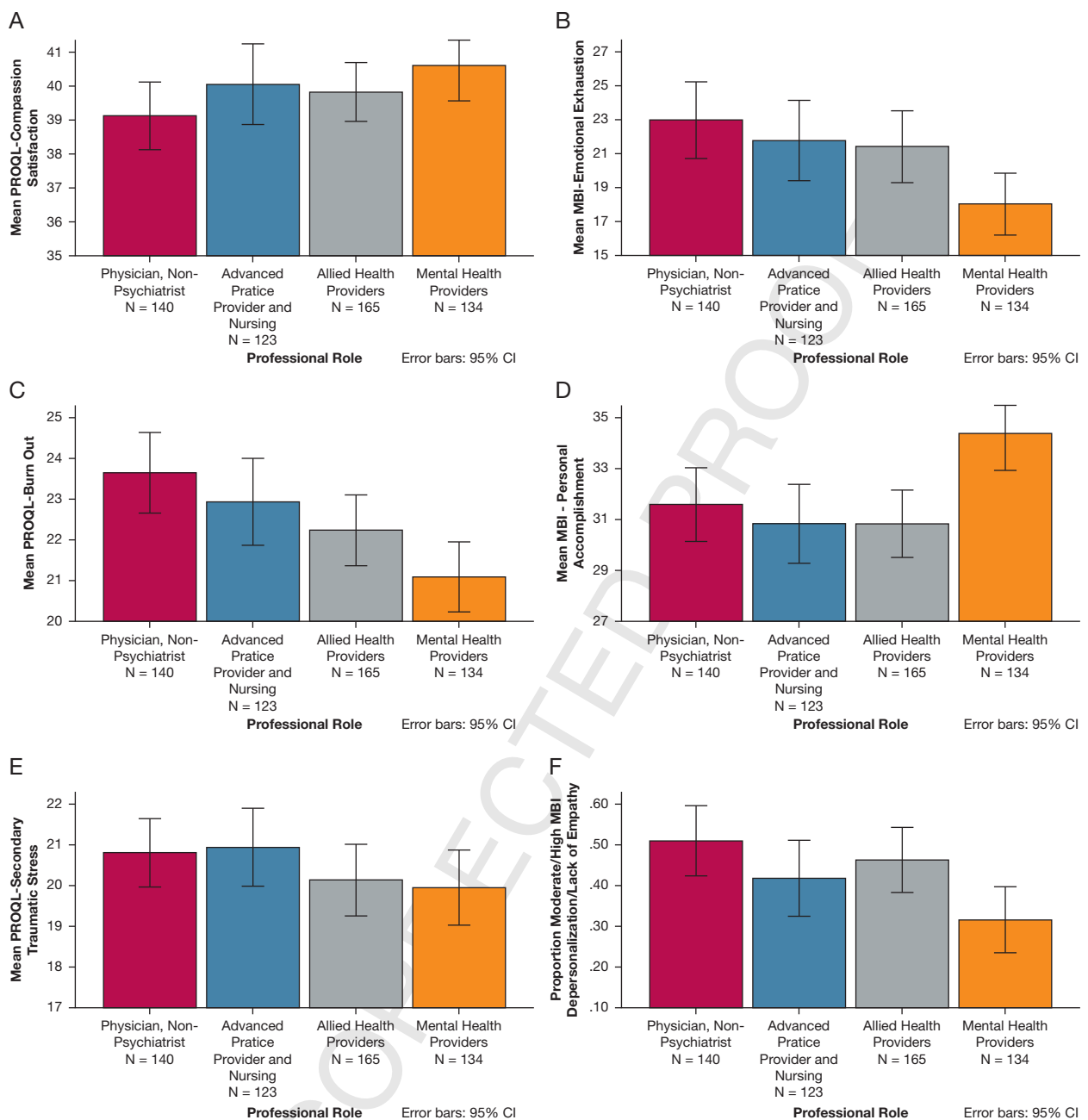


Figure 1 – A-F, ProQOL (A-C) and MBI (D-F) subscales by professional role. The range of each y axis represents 1 SD in each outcome. The scores represented on each y axis were then selected to demonstrate differences between groups while attempting to keep subscale scores represented among the same scales as consistent as possible. Error bars represent 95% CIs. Scoring criteria included the following: ProQOL (compassion satisfaction, burnout, and secondary traumatic stress) (low: ≤ 22 ; moderate: 23-41; high: ≥ 42), MBI emotional exhaustion (low: ≤ 17 ; moderate: 18-29; high: ≥ 30), MBI personal accomplishment (low: ≤ 33 ; moderate: 34-39; high: ≥ 40), and MBI depersonalization (low: ≤ 5 ; moderate: 6-11; high: ≥ 12). MBI = Maslach Burnout Inventory; ProQOL = Professional Quality of Life Scale.

Multivariable Models Evaluating Associations Between Professional Role, Demographics, and Outcomes

When adjusted for other provider characteristics, professional role remained significantly associated with all measures of burnout (Table 3). Mental health

providers endorsed significantly lower scores than physicians and APPs/nurses on both the MBI-burnout and MBI-emotional exhaustion scales. Mental health providers had higher MBI personal accomplishment scores and lower MBI depersonalization scores than all other providers. Notably, the odds of having a

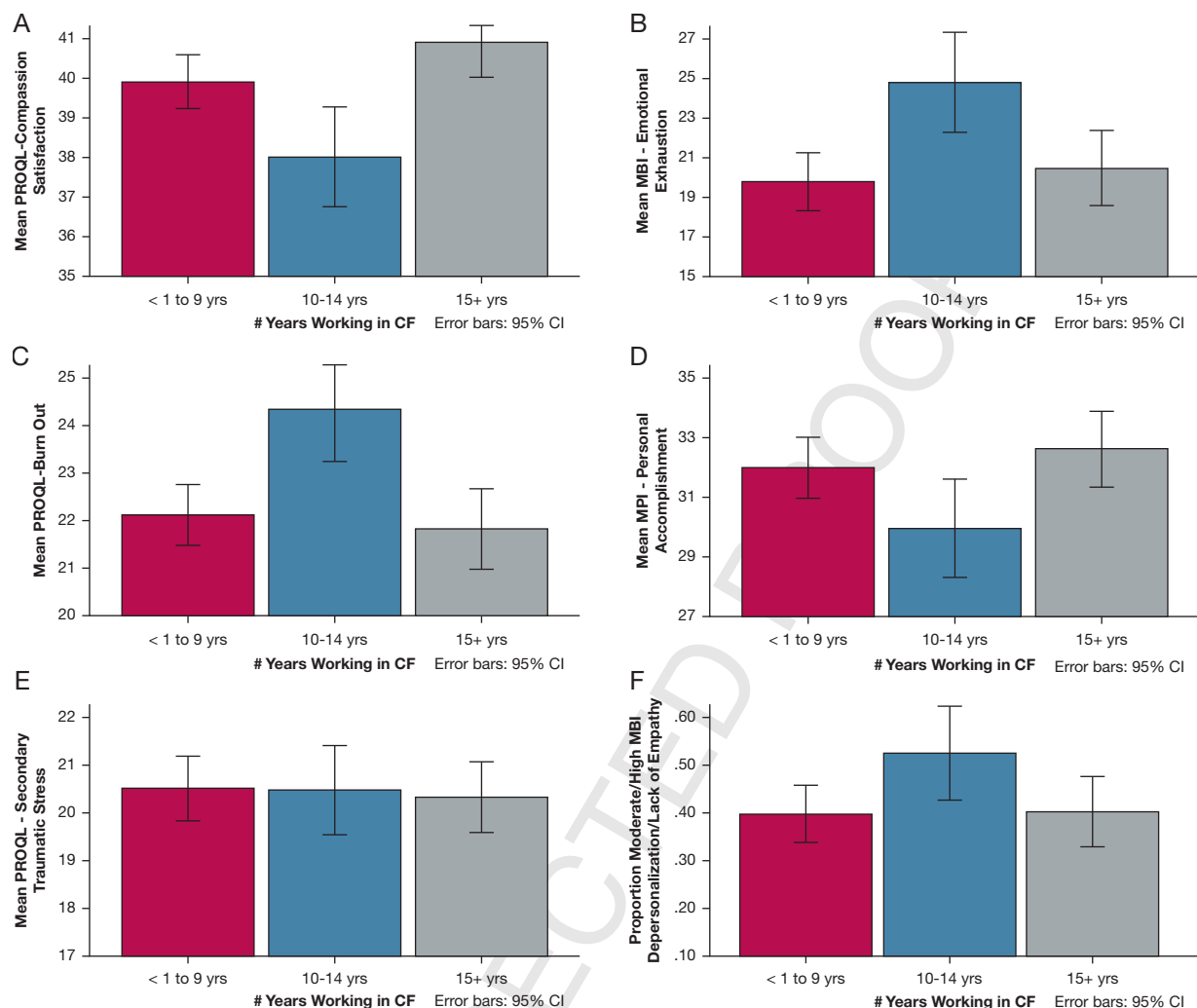


Figure 2 – A-F, ProQOL (A-C) and MBI (D-F) subscales by number of years working in CF. The range of each y axis represents 1 SD in each outcome. The scores represented on each y axis were then selected to demonstrate differences between groups while attempting to keep subscale scores represented among the same scales as consistent as possible. Error bars represent 95% CIs. Scoring criteria included the following: ProQOL (compassion satisfaction, burnout, and secondary traumatic stress) (low: ≤ 22 ; moderate: 23-41; high: ≥ 42), MBI emotional exhaustion (low: ≤ 17 ; moderate: 18-29; high: ≥ 30), MBI personal accomplishment (low: ≤ 33 ; moderate: 34-39; high: ≥ 40), and MBI depersonalization (low: ≤ 5 ; moderate: 6-11; high: ≥ 12). CF = cystic fibrosis; MBI = Maslach Burnout Inventory; ProQOL = Professional Quality of Life Scale.

moderate/high depersonalization score were almost doubled for APPs/nurses and allied health providers, and almost tripled for physicians.

After adjustment for other provider characteristics, women (vs men) continued to demonstrate lower MBI personal accomplishment scores, and Latino (vs non-Latino) participants reported higher ProQOL burnout and MBI emotional exhaustion, and lower MBI personal accomplishment. When adjusted for other provider characteristics, working with CF for ≥ 15 (vs 10-14) years and higher age both remained associated with lower ProQOL burnout and MBI emotional exhaustion scores. When adjusted, working with a

pediatric vs adult patient population was no longer associated with any of the outcome measures. Size of CF program was not associated with the outcome measures.

Relationships Among Measures of Burnout

We expected to find significant, inverse correlations between positive aspects of professional satisfaction (ProQOL compassion satisfaction) and sense of accomplishment (MBI personal accomplishment) and measures of burnout (ProQOL burnout, MBI emotional exhaustion, MBI depersonalization). Associations among the ProQOL and MBI scales were moderate to large, significant, and in the hypothesized directions (e-Table 2).

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TABLE 3] Adjusted Models Evaluating Professional Characteristics by ProQOL BO, MBI EE, MPI PA, and MPI DP

Provider Characteristics	ProQOL BO (n = 541), Estimate (95% CI)	MBI EE (n = 516), Estimate (95% CI)	MBI PA (n = 501), Estimate (95% CI)	MBI DP Moderate/High (n = 556), OR (95% CI) ^a
Professional role				
Physician, nonpsychiatrist	2.4 (0.9 to 3.9)	3.9 (0.7 to 7.2)	−3.9 (−6.3 to −1.6)	2.8 (1.7 to 4.9)
Advanced practice provider and nursing	2.0 (0.6 to 3.5)	3.8 (0.7 to 7.1)	−4.2 (−6.4 to 2.0)	1.8 (1.01 to 3.0)
Allied health providers	0.7 (−0.7 to 2.1)	1.8 (−1.2 to 4.8)	−4.0 (−6.0 to −2.0)	1.8 (1.1 to 3.0)
Mental health providers (Ref)	(Ref)	(Ref)	(Ref)	(Ref)
Age	−0.9 (−1.4 to −0.4)	−2.8 (−4.0 to −1.7)	0.6 (−0.2 to 1.3)	0.7 (0.6 to 0.8)
Women (vs men)	−	−	−2.7 (−4.8 to −0.6)	−
Latino (vs non-Latino)	2.2 (0.5 to 3.8)	5.4 (1.5 to 9.2)	−2.9 (−5.5 to −0.2)	−
Years working with CF				
< 10	−1.6 (−3.1 to −0.9)	−1.0 (−4.2 to 2.2)	1.5 (−0.7 to 3.7)	0.6 (0.4 to 1.02)
≥ 15	−2.2 (−3.5 to −0.9)	−5.2 (−8.2 to −2.2)	1.8 (−0.2 to 3.8)	0.7 (0.4 to 1.1)
10-14 (Ref)	(Ref)	(Ref)	(Ref)	(Ref)
Patient population				
Pediatric	0.6 (−0.7 to 1.8)	−	−	−
Adult	1.1 (0.3 to 2.4)	−	−	−
Pediatric and adult (Ref)	(Ref)	−	−	−

Boldface parameters are significant when adjusted for all other variables in the model (professional role, age, gender, ethnicity, years working with CF, and patient population). Scoring criteria included the following: ProQOL BO (low: ≤ 22; moderate: 23-41; high: ≥ 42), MBI EE (low: ≤ 17; moderate: 18-29; high: ≥ 30), MBI PA (low: ≤ 33; moderate: 34-39; high: ≥ 40), and MBI DP (low: ≤ 5; moderate: 6-11; high: ≥ 12). BO = burnout; CF = cystic fibrosis; DP = depersonalization; EE = emotional exhaustion; MBI = Maslach Burnout Inventory; PA = personal accomplishment; ProQOL = Professional Quality of Life Scale; Ref = reference.

^aFor MBI DP, ORs and 95% CIs are reported.

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Contributors to Burnout

The top 5 most frequently endorsed contributors to burnout were rank-ordered across providers, including (1) administrative tasks (80.0%), (2) documentation (79.4%), (3) staffing (74.6%), (4) time with patients (73.4%), and (5) responsibility (72.7%) (e-Table 3, Fig 3). Other top contributors impacting burnout a lot were compensation and lack of respect from administration.

Contributors to burnout differed by professional role. Compared with other roles, physicians more often rated

the electronic health record, documentation, administrative tasks, too many work hours, and working outside of business hours as contributing to burnout a lot. Compared with other professions, mental health providers more often reported that excessive regulations contributed to burnout a lot (e-Table 3, Fig 3).

All contributors to burnout except trauma exposure (infrequently endorsed) were associated with all study outcomes at $P < .008$ (Bonferroni correction) (e-Table 4). Excessive responsibility and too many work hours had large relationships with burnout on the MBI

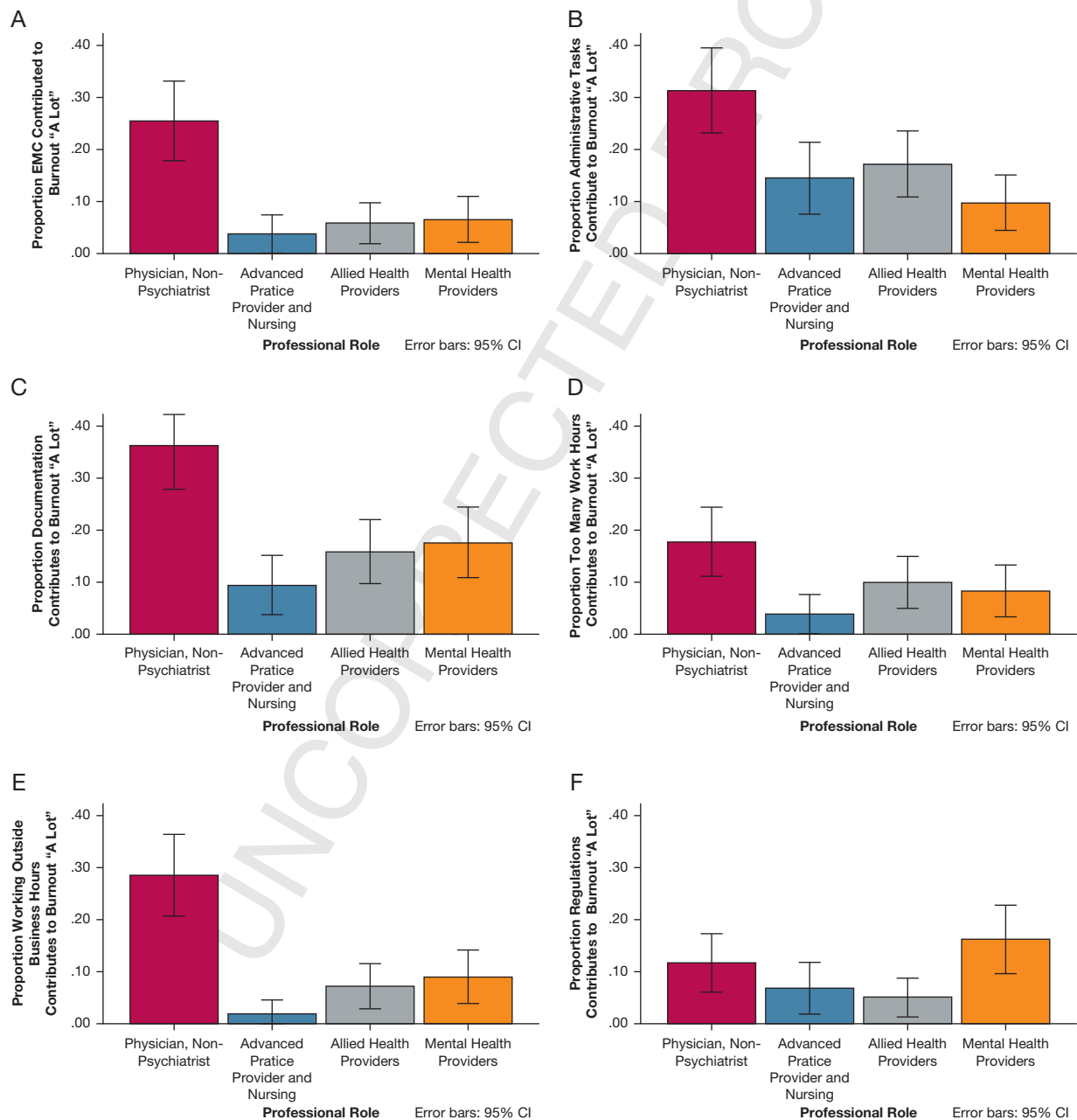


Figure 3 – A-F, Contributors to burnout by professional role. Error bars represent 95% CIs.

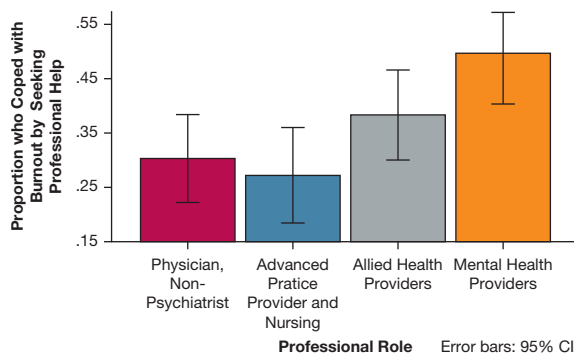


Figure 4 – Seeking professional help to cope with burnout by professional role. Error bars represent 95% CIs.

emotional exhaustion subscale ($r = 0.51$ and 0.52 , respectively; $P < .001$).

Coping With Burnout

Talking with family or friends was the most commonly endorsed method of coping with burnout (96.0%) (e-Fig 1). Across respondents, 36.8% reported ever seeking professional help to cope with burnout; however, this result differed by professional role ($\chi^2 = 14.70$; $P = .002$) (Fig 4); almost one-half (49.6%) of mental health providers had sought professional help to cope with burnout compared with less than one-third of physicians (30.2%) and APPs/nurses (27.2%). When physicians were divided by gender, significantly more female (40%) than male physicians (19%) had sought professional help ($\chi^2 = 5.5$ continuity correction; $P = .019$).

Discussion

To our knowledge, this is the first national study of burnout and professional QOL in multidisciplinary CF health care providers. Our results documented relatively high ratings of burnout and low ratings of personal accomplishment. We do not know if these ratings would be worse without the team-based care model; however, our scores indicate that burnout is moderately high and personal accomplishment is low. Two published abstracts asked physicians working on CF teams, some of whom were trainees, about their levels of burnout and plans to specialize in CF care.^{34,35} Rates of burnout were reported by 34% to 37% of physician respondents.^{34,35}

Although we expected providers to rate satisfaction with helping others and their sense of personal accomplishment in the high range, compassion satisfaction was moderate and personal accomplishment was low. Burnout on the ProQOL was low to moderate,

and on the MBI, respondents reported moderate emotional exhaustion. As expected, men reported higher personal accomplishment than women, and Hispanic providers reported higher burnout, secondary traumatic stress, and emotional exhaustion and lower personal accomplishment than non-Hispanic, White providers.

Contrary to expectations, years working in CF was curvilinear, with those new to CF and those who have worked in CF ≥ 15 years reporting better outcomes across scales than those in the 10- to 14-year category. This may be a survivor effect, with providers leaving CF care midcareer if they are burned out or dissatisfied, whereas others who find working in CF to be fulfilling tend to stay in the field longer. Alternatively, this U-shaped pattern has also been found in classic studies of happiness and well-being.³⁶ Happiness is robust early in life, dips down in midlife, and then bounces back later in life. Blanchflower confirmed this pattern in 145 countries.³⁶

Our predictions that physicians, who are in higher status positions with greater autonomy, would report the highest professional satisfaction and lowest burnout were not supported. In comparison with mental health professionals, physicians, nurses, and allied health providers reported significantly lower personal accomplishment and higher burnout. Furthermore, compared with mental health providers, physicians reported higher emotional exhaustion, and both physicians and allied health providers reported higher depersonalization. In comparison with psychologists/psychiatrists, social workers reported higher burnout and secondary traumatic stress and lower ratings of personal accomplishment. These data are important in considering how to retain social workers and other professionals on CF teams.

Rates of burnout and turnover among physicians and other providers have been termed a hidden health care crisis, particularly after the COVID-19 pandemic.¹² Efforts to address burnout, particularly among physicians, have included individual-level approaches such as increasing awareness, assessment of burnout, and providing wellness interventions. It is possible that satisfaction with helping others and a sense of personal accomplishment may mitigate the effects of burnout; we found strong inverse relationships between these constructs. In addition, our results suggested that working outside business hours, burdensome documentation, and administrative tasks were major

contributors to burnout, which argue for making organizational and systemic changes to health care systems.³⁷⁻³⁹

In this sample, the most commonly endorsed strategy for coping with burnout was talking to family and friends. Notably, physicians and APPs/nurses were significantly less likely to endorse seeking professional help than mental health providers, who consistently reported lower rates of burnout than other providers in this study. Among physicians, more female than male providers sought professional help, highlighting the need to evaluate the feasibility and acceptability of interventions across demographic groups. Supportive interventions should include education regarding the neurobiology of stress and burnout,¹⁴ and organizing topic-focused peer discussion groups.⁴⁰ Hente et al⁴¹ offered a 6-hour mindfulness-based cognitive therapy training to improve stress and burnout, held during work hours, to 56 multidisciplinary CF care team members; 43% participated, 83% of whom were female. The intervention improved well-being among participants, including mood, anxiety, perceived stress, and depersonalization; the authors emphasized that such trainings should be considered voluntary rather than mandatory.⁴¹

The American Medical Association and other professional organizations have called for structural and legislative reforms to reduce barriers to seeking help, including discriminatory practices that affect physician licensing and credentialing in many states.⁴² As of May 2025, 37 medical licensure boards and 635 hospitals have verified that their licensing and credentialing boards are free from intrusive mental health questions and stigmatizing language, a substantial increase from 2022 when only 22 boards and health systems had modified their procedures.⁴³

Systemic contributors to burnout in our study echoed those commonly identified in the literature.^{2,15,44} Physicians more often rated documentation, administrative tasks, too many work hours, and working outside of business hours as contributing to burnout a lot, whereas mental health clinicians more often rated excessive regulations as contributing a lot.^{2,15,44} Physician training models, which include long hours (eg, 80-hour weeks for residents), low rates of clinical interactions (eg, residents report only 12% of their time spent with patients), and high administrative loads, have previously been shown to contribute to burnout.⁴⁵ Several initiatives by the American Medical

Association and others are aimed at streamlining administrative burden, providing alternative scheduling models for residents, and increasing time spent in patient interactions (eg, Back to Bedside initiatives).⁴⁶ The drivers of burnout are optimally addressed at multiple levels, including the individual, care team, institutional, and national contexts.^{47,48}

Limitations

Our response rate was lower than anticipated. However, sample sizes were similar across professional roles and thus not likely to impact comparisons of roles. In addition, low response rates in large-scale burnout surveys are very common. Simonetti et al⁴⁹ evaluated nonresponse bias and its effects on rates of burnout in 27,226 primary care providers using response propensity scores in a mixed-effects logistic regression and found no significant differences in rates of burnout. They concluded that workforce surveys of burnout yield valid estimates even in the context of low response rates. Two other studies used intensive follow-up of nonresponders to evaluate their levels of burnout and found no significant differences in rates of burnout between responders and nonresponders.^{50,51}

We also strengthened the measurement of burnout by administering 2 standardized measures in their entirety. Additionally, the respondents across professions were primarily female, which converges with data on the feminization of health care.⁵² Note, however, that nearly one-half of the physicians were male. Although pulmonology as a specialty includes predominantly male providers,⁵³ pediatric specialties, such as pediatric pulmonology, have a higher proportion of female providers.⁵⁴

Interpretation

In one of the few national studies of health care provider burnout in the context of team-based care, participants reported moderate rates of burnout and low rates of professional satisfaction. In contrast with mental health providers, physicians reported the highest rates of emotional exhaustion and depersonalization, placing them at greatest risk for burnout. Physicians were also the least likely to seek professional support, perhaps due to administrative and structural barriers. In addition to promoting individual well-being, systems-level initiatives are critical to support multidisciplinary team members providing care for people with chronic pulmonary disease.

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1373 References

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